An interpolation family for atomic–absolutely continuous ball discrepancy

Candidate partial answer to the sharpness question in arXiv:2310.10120

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Status: candidate partial result, with a full literal-weight observation. For equal weights and a probability density of fixed L^p norm, the optimal N-power is $N^{-1-1/d}$ for every $1 < p < 2$, not the power $N^{-1-q/(2d)}$ in the source lower bound. More generally, a continuum of rigorous lower bounds interpolates between these two endpoints. This does not settle the optimal joint dependence on N and $\|f\|_p$. If the source theorem is read literally with arbitrary nonnegative weights, however, its full displayed dependence is sharp by a one-active-weight construction.

1 The source question

Let $1 < p < 2$, let $q = p/(p-1)$, and let

$$\mu_N = \frac{1}{N}\sum_{j=1}^{N} \delta_{z_j}, \qquad \nu = \mu_N - f(x)\,dx,$$

where $f \geq 0$ and $\int_{\mathbb{T}^d} f = 1$. For fixed $0 < a < b < 1/2$, put

$$\mathcal{E}_{a,b}(\nu) = \int_a^b \int_{\mathbb{T}^d} |\mathcal{D}_N(x,r)|^2\,dx\,dr.$$

The Fourier computation in Brandolini–Colzani–Travaglini [1] gives

$$\mathcal{E}_{a,b}(\nu) \asymp_{a,b,d} \sum_{m\in\mathbb{Z}^d} (1+|m|)^{-d-1}|\widehat{\nu}(m)|^2 = \|\nu\|^2_{H^{-(d+1)/2}(\mathbb{T}^d)}. \tag{1}$$

Their Theorem 6 proves

$$\mathcal{E}_{a,b}(\nu) \gtrsim N^{-1-q/(2d)}\|f\|_p^{-q/d} \tag{2}$$

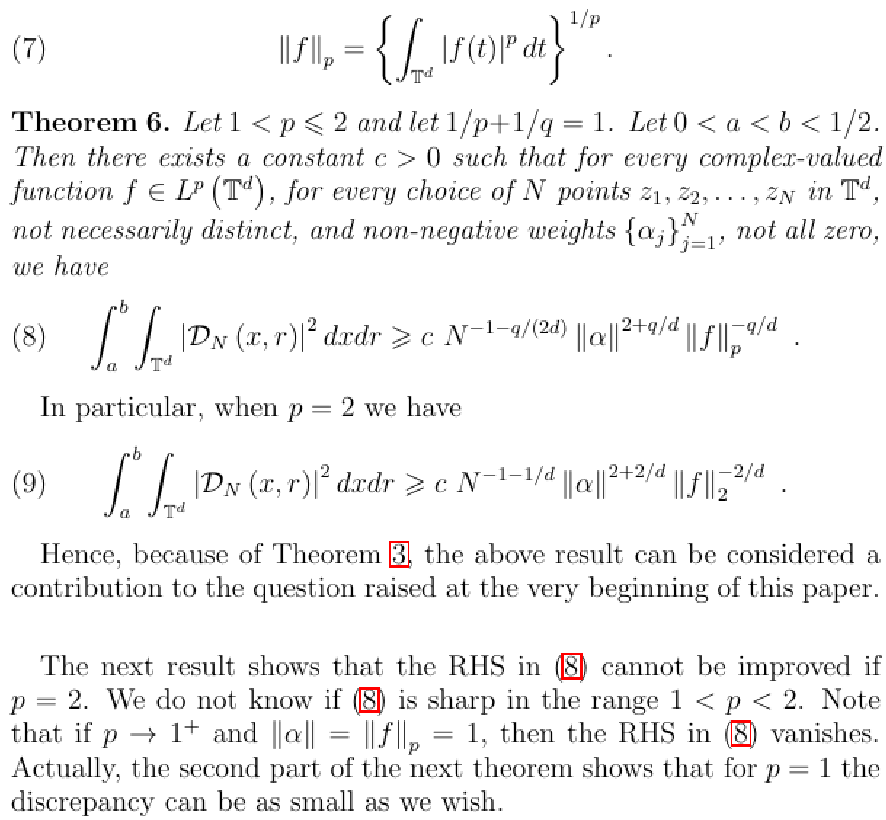
in the equal-weight normalization, and asks whether this is sharp for $1 < p < 2$. The exact statement is reproduced in Figure 1.

2 The interpolation theorem

For $u \in [p,2]$ define

$$\theta_u = 1 - \frac{1}{u}, \qquad \beta_u = d + 1 - 2d\theta_u = 1 + d\left(\frac{2}{u} - 1\right).$$

Notice that $\beta_u \geq 1$.



$$\|f\|_p = \left\{ \int_{\mathbb{T}^d} |f(t)|^p \, dt \right\}^{1/p} . \tag{7}$$

Theorem 6. *Let $1 < p \leqslant 2$ and let $1/p + 1/q = 1$. Let $0 < a < b < 1/2$. Then there exists a constant $c > 0$ such that for every complex-valued function $f \in L^p\left(\mathbb{T}^d\right)$, for every choice of N points $z_1, z_2, \ldots, z_N$ in $\mathbb{T}^d$, not necessarily distinct, and non-negative weights $\{\alpha_j\}_{j=1}^N$, not all zero, we have*

$$\int_a^b \int_{\mathbb{T}^d} |\mathcal{D}_N(x,r)|^2 \, dx dr \geqslant c\; N^{-1-q/(2d)} \, \|\alpha\|^{2+q/d} \, \|f\|_p^{-q/d} \; . \tag{8}$$

In particular, when $p = 2$ we have

$$\int_a^b \int_{\mathbb{T}^d} |\mathcal{D}_N(x,r)|^2 \, dx dr \geqslant c\; N^{-1-1/d} \, \|\alpha\|^{2+2/d} \, \|f\|_2^{-2/d} \; . \tag{9}$$

Hence, because of Theorem 3, the above result can be considered a contribution to the question raised at the very beginning of this paper.

The next result shows that the RHS in (8) cannot be improved if $p = 2$. We do not know if (8) is sharp in the range $1 < p < 2$. Note that if $p \to 1^+$ and $\|\alpha\| = \|f\|_p = 1$, then the RHS in (8) vanishes. Actually, the second part of the next theorem shows that for $p = 1$ the discrepancy can be as small as we wish.

Figure 1: Theorem 6 and the sharpness question in the source preprint, p. 5.

Theorem 1 (Equal-weight interpolation family)**.** *Let $f \geq 0$, $\int_{\mathbb{T}^d} f = 1$, and $F = \|f\|_p$. For every $u \in [p, 2]$ and every equal-weight empirical measure μ_N,*

$$\mathcal{E}_{a,b}(\mu_N - f\,dx) \geq c_{a,b,d,p,u}\, N^{-2\theta_u - q\theta_u\beta_u/d} F^{-q\beta_u/d}. \tag{3}$$

At $u = 2$, one has $\theta_u = 1/2$ and $\beta_u = 1$, so (3) is exactly (2). At the other endpoint, $u = p$, it becomes

$$\mathcal{E}_{a,b}(\mu_N - f\,dx) \geq c_{a,b,d,p} N^{-1-1/d} F^{-(q-2)-q/d}. \tag{4}$$

Proof intuition

Smooth the empirical measure at a scale h. Positivity and equal weights force the L^u norm of the resulting sum of N bumps to be at least $N^{-1/u'} h^{-d/u'}$, regardless of collisions among the points. Young's inequality bounds the smoothed density by $F h^{-d(1/p-1/u)}$. The difference of the two exponents of h is exactly $1/q$. Choosing the scale at which the atomic term is twice the density term leaves a definite L^u, hence L^2, discrepancy. The critical Sobolev multiplier costs $h^{-(d+1)}$. Varying u gives the entire family.

Proof. Choose $\phi \in C_c^\infty(\mathbb{R}^d)$ nonnegative, with $\int \phi = 1$, and periodize

$$\phi_h(x) = h^{-d} \phi(x/h), \qquad 0 < h < h_0.$$

Put $g_h = \mu_N * \phi_h$. Since $(\sum_j a_j)^u \geq \sum_j a_j^u$ for nonnegative a_j,

$$\|g_h\|_u^u = \int_{\mathbb{T}^d} \left| \frac{1}{N} \sum_{j=1}^{N} \phi_h(x - z_j) \right|^u dx \tag{5}$$

$$\geq N^{-u} \sum_{j=1}^{N} \|\phi_h\|_u^u = c_{\phi,u} N^{1-u} h^{-d(u-1)}. \tag{6}$$

Thus

$$\|g_h\|_u \geq c_0 N^{-\theta_u} h^{-d\theta_u}. \tag{7}$$

Let v be determined by $1 + 1/u = 1/p + 1/v$. Young's inequality gives

$$\|f * \phi_h\|_u \leq \|f\|_p \|\phi_h\|_v \leq C_0 F h^{-d(1/p - 1/u)}. \tag{8}$$

Because

$$\theta_u - \left(\frac{1}{p} - \frac{1}{u}\right) = 1 - \frac{1}{p} = \frac{1}{q},$$

we may choose

$$h = c_1 N^{-q\theta_u/d} F^{-q/d}, \tag{9}$$

with $c_1 = c_1(d, p, u, \phi) > 0$ small enough that the right-hand side of (7) is at least twice that of (8). Since $F \geq 1$, the scale lies below a fixed h_0 after decreasing c_1 if necessary. Therefore

$$\|(\mu_N - f\,dx) * \phi_h\|_u \geq c_2 N^{-\theta_u} h^{-d\theta_u}. \tag{10}$$

As $u \leq 2$ and $|\mathbb{T}^d| = 1$, the same lower bound holds for the L^2 norm.

The rapid decay of $\widehat{\phi}$ implies, uniformly for $m \in \mathbb{Z}^d$,

$$|\widehat{\phi}(hm)|^2 \leq C h^{-d-1} (1 + |m|)^{-d-1}.$$

Parseval consequently gives

$$\|(\mu_N - f\,dx) * \phi_h\|_2^2 \leq C h^{-d-1} \|\mu_N - f\,dx\|_{H^{-(d+1)/2}}^2. \tag{11}$$

Combining (10)–(11), then substituting (9), yields

$$\|\mu_N - f\,dx\|_{H^{-(d+1)/2}}^2 \geq c N^{-2\theta_u} h^{d+1-2d\theta_u} = c N^{-2\theta_u - q\theta_u\beta_u/d} F^{-q\beta_u/d}.$$

Equation (1) finishes the proof. □

3 A sharp fixed-norm consequence

Corollary 1 (Optimal N-power when F is fixed)**.** *Fix $F \geq 1$. Among probability densities with $\|f\|_p = F$ and equal-weight N-point measures, the best possible N-power in the squared averaged-ball discrepancy is $N^{-1-1/d}$ (with constants allowed to depend on F). The upper bound holds for $N = H^d$ and hence already proves sharpness of the power.*

Proof. The lower bound follows from (4). For the upper bound, let $Q \subset \mathbb{T}^d$ be a cube of volume F^{-q} and set $f = F^q \mathbf{1}_Q$, so $\int f = 1$ and $\|f\|_p = F$. If $N = H^d$, partition Q into N congruent subcubes Q_j of diameter

$$\ell \asymp F^{-q/d} N^{-1/d}.$$

Choose independent X_j uniformly in Q_j. In the Hilbert space $H^{-(d+1)/2}(\mathbb{T}^d)$ the centered random measures

$$Y_j = \delta_{X_j} - N f \mathbf{1}_{Q_j}\, dx$$

have vanishing cross terms. Moreover,

$$\|\delta_x - \delta_y\|^2_{H^{-(d+1)/2}} \leq C_d |x-y|. \tag{12}$$

Indeed, split the defining Fourier sum at $|m| = |x-y|^{-1}$, use $|e^{2\pi i m x} - e^{2\pi i m y}| \leq C|m||x-y|$ below the split, and the trivial bound above it. Hence

$$\mathbb{E}\left\|\frac{1}{N}\sum_{j=1}^N Y_j\right\|^2_{H^{-(d+1)/2}} = \frac{1}{N^2}\sum_{j=1}^N \mathbb{E}\|Y_j\|^2_{H^{-(d+1)/2}} \leq C_d \frac{\ell}{N}.$$

Some realization therefore satisfies

$$\mathcal{E}_{a,b}(\mu_N - f\,dx) \lesssim N^{-1-1/d} F^{-q/d},$$

by (1). $\square$

This fixed-norm rate is consistent with the independent May 2026 theorem of Colasanto–Focardi–Fornasier–Mattesini [2]: for the power kernel $-|x-y|$ and a d-Ahlfors regular target, their energy distance squared has sharp rate $N^{-1-1/d}$. Their theorem does not cover arbitrary L^p densities or determine the dependence on F.

4 The literal arbitrary-weight statement is sharp

The wording surrounding the source question points to equal weights, as does the construction in its Theorem 7. The displayed lower bound itself permits arbitrary nonnegative weights. Under that literal reading there is a short complete answer.

Proposition 1 (One-active-weight sharpness)**.** *For every $1 < p \leq 2$, the joint powers of N, $\|\alpha\|$, and $\|f\|_p$ in the source inequality are simultaneously sharp when arbitrary nonnegative weights are allowed.*

Proof. Take $\alpha_1 = \sqrt{N}$, $\alpha_2 = \cdots = \alpha_N = 0$ and $z_1 = 0$. Then the normalized weight norm in the source is $\|\alpha\| = 1$, while the atomic measure is $N^{-1/2}\delta_0$. With the same approximate identity as above, set

$$f_h = N^{-1/2}\phi_h.$$

Then $\|f_h\|_p \asymp N^{-1/2} h^{-d/q}$ and, as $h \downarrow 0$,

$$\mathcal{E}_{a,b}(N^{-1/2}(\delta_0 - \phi_h dx)) \asymp N^{-1} h.$$

The last equivalence follows directly by splitting the Fourier series in (1) at $|m| = h^{-1}$; the tail already has size h. On the other hand, the right-hand side of the source bound equals

$$N^{-1-q/(2d)}\|\alpha\|^{2+q/d}\|f_h\|_p^{-q/d} \asymp N^{-1-q/(2d)} N^{q/(2d)} h = N^{-1}h.$$

Thus all displayed exponents are attained. $\square$

5 What remains open

The intended equal-weight problem asks for the optimal joint dependence on N and $F = \|f\|_p$. Theorem 1 supplies a continuous tradeoff, but its $u = p$ endpoint has the extra factor $F^{-(q-2)}$, while the source $u = 2$ endpoint has the weaker N-power. A natural attempt to remove this gap uses empty-ball or Whitney bump tests at scales adapted to the local L^p mass. At the critical Sobolev order each bump has exactly the right scale, but an arbitrary density may distribute its mass across infinitely many spatial scales; the resulting stopping-time sum is not controlled by $\|f\|_p$ alone. No valid argument found here removes that multiscale obstruction, and no claim about the optimal joint exponents is made.

References

[1] L. Brandolini, L. Colzani, and G. Travaglini, *Discrepancy and approximation of absolutely continuous measures with atomic measures*, J. Geom. Anal. **35** (2025), Paper 97; arXiv:2310.10120.

[2] F. Colasanto, M. Focardi, M. Fornasier, and F. Mattesini, *Sharp Rates of MMD Empirical Estimation with Power Kernels*, arXiv:2605.18497v2 (2026).